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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

IMPORTANCE OF ELECTRICAL CIRCUIT COURSE FOR COMPUTER SCIENCE ENGINEERS

Connecting circuit fundamentals with today's computing technologies

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1. Introduction

Computer Science and Engineering (CSE) is often associated with programming, software development, algorithms, and data processing. However, every program ever written ultimately runs on a physical machine made of electronic components. Those components need electrical power to operate, they exchange information as electrical signals, and they perform every operation through the controlled movement of electric charge. A line of code such as **a = b + c** is not just an abstract idea; it becomes a sequence of voltage changes that travel through wires and transistors inside the processor.

A circuit-level understanding helps a CSE student see how software instructions become real actions inside a computer. The Electrical Circuit course introduces the language needed for this: voltage, current, resistance, power, energy, and the behaviour of simple networks of components. These ideas are not limited to traditional electrical engineering. They are directly useful in computer architecture, digital electronics, embedded systems, robotics, artificial intelligence (AI), the Internet of Things (IoT), electric vehicles, and renewable energy.

This report explains the basic concepts of electrical circuits, shows through examples how they connect to modern computing, and argues that the Electrical Circuit course is a foundation for a CSE student, not a separate or optional subject.

> **Key idea**
> Software decides *what* a computer should do; electrical circuits decide *how* it physically happens, how fast it happens, and how much energy it costs.

2. Basic Concepts of Electrical Circuits

An electrical circuit is a closed, connected path through which electric current can flow. A basic circuit contains four parts: a **source** of electrical energy (battery or power supply), **conductors** (wires or copper tracks), a **switch or control element**, and a **load** (an LED, a motor, a resistor, or a chip) that uses the energy. If the path is broken anywhere, current stops, which is exactly how a switch turns a device on and off.

2.1 Fundamental Quantities

Quantity	Symbol	Unit	Meaning
Voltage	V	volt (V)	Electrical potential difference that pushes charge through a circuit
Current	I	ampere (A)	Rate of flow of electric charge
Resistance	R	ohm (Ω)	Opposition to the flow of current
Power	P	watt (W)	Rate at which electrical energy is transferred or used
Energy	E	joule (J)	Total work done or energy consumed over time
Capacitance	C	farad (F)	Ability of a component to store charge in an electric field

2.2 Ohm’s Law and Electrical Power

The most important relationship in basic circuit analysis is Ohm’s Law, which states that the voltage across a resistor is proportional to the current through it:

$$V = I \times R$$

where V is the voltage in volts, I is the current in amperes, and R is the resistance in ohms. Electrical power follows from voltage and current:

$$P = V \times I = I^2R = V^2/R$$

These formulas explain a very practical fact: a component that carries more current dissipates more power, and that power appears as heat. This is why processors need heat sinks and fans, why phone chargers become warm, and why a wire that is too thin can overheat.

2.3 Worked Example: Protecting an LED

Suppose a 5 V microcontroller pin must light an LED whose forward voltage is about 2 V and which should carry roughly 10 mA. The resistor must drop the remaining 3 V:

$$R = (5\ V - 2\ V) / 0.010\ A = 300\ \Omega$$

The nearest common value is 330 Ω, giving a current of 3 V / 330 Ω ≈ 9.1 mA. The resistor dissipates $P = I^2R \approx (0.0091)^2 \times 330 \approx 27$ mW, which a small quarter-watt resistor handles easily. Without the resistor, the LED would draw excessive current and burn out. This simple calculation is the kind of reasoning every embedded-systems student uses on the first day with a development board.

2.4 Series and Parallel Circuits

- **Series:** components share the same current, and resistances add: $R_{total} = R_1 + R_2 + \dots$ Voltage divides between them.
- **Parallel:** components share the same voltage, and currents add. For two resistors, $R_{total} = (R_1R_2)/(R_1 + R_2)$; two 100 Ω resistors in parallel give 50 Ω.
- **Voltage divider:** two resistors in series produce a fraction of the input voltage: $V_{out} = V_{in} \times R_2/(R_1 + R_2)$. With $V_{in} = 5$ V and $R_1 = R_2 = 10$ kΩ, $V_{out} = 2.5$ V. Many analogue sensors use this idea.

2.5 Kirchhoff’s Laws, DC and AC

Kirchhoff's Current Law (KCL) says that the total current entering a node equals the total current leaving it. **Kirchhoff's Voltage Law (KVL)** says that the voltages around any closed loop add up to zero. Together with Ohm's Law, they allow us to analyse almost any resistive network. Circuits can be powered by **direct current (DC)**, where voltage has a constant polarity (batteries, USB, chip supply rails), or **alternating current (AC)**, where voltage varies periodically (mains electricity). Computers convert AC from the wall into low-voltage DC with a power supply, because digital chips need steady DC rails such as 5 V, 3.3 V, or about 1 V.

2.6 Capacitors and Inductors

Besides resistors, circuits use **capacitors**, which store energy in an electric field, and **inductors**, which store energy in a magnetic field. In computers, capacitors placed near a processor smooth out sudden changes in supply voltage (decoupling), and the same principle of charge storage is used in DRAM memory cells. Inductors appear in the switching power converters that supply the CPU and charge laptop batteries. Understanding them explains why signals take time to change and why circuits cannot switch instantly.

CSE students do not need to become circuit-design specialists to benefit from these ideas. Knowing the basics helps them understand why electronic devices require suitable power, why components heat up, why signals can become noisy, and why a circuit must be designed carefully.

3. Why Electrical Circuits Matter to CSE Engineers

Electrical circuits connect the abstract world of software with the physical world of computing. A program contains instructions and data, but a processor uses electronic signals to carry out those instructions. Circuit knowledge helps students understand how information is represented, how components communicate, and how a computer's performance is affected by power and hardware design. The knowledge is useful in several CSE areas:

CSE area	How circuit knowledge helps	Example
Computer Architecture	Explains how processors, memory, and buses exchange signals, and why clock speed and power are limited	Why a CPU slows down when it overheats
Digital Electronics	Shows how binary values (0 and 1) are represented by voltage levels and processed by logic gates	Designing an adder or counter
Embedded Systems	Connects software with sensors, controllers, and output devices safely	Reading a sensor with a microcontroller
Computer Networks	Explains signal transmission, noise, and attenuation in cables and wireless links	Why long cables lose signal quality
Troubleshooting	Helps recognise power, connection, grounding, and signal problems quickly	A board that will not power on
Green Computing	Links hardware design to energy use and heat in data centres and devices	Reducing the power cost of servers

In short, a software engineer who understands circuits can write code that respects hardware limits, debug problems that are not caused by the code at all, and communicate clearly with electronics and hardware teams.

4. Electrical Circuits in Modern Computer Hardware

A computer's central processing unit (CPU) contains electronic circuits that perform arithmetic, logical, and control operations. Memory circuits store instructions and data so that the processor can access them quickly. The motherboard provides copper pathways for power and signals between the CPU, memory, storage, and other components. A graphics processing unit (GPU) contains thousands of simple processing units and supporting circuits that can handle large numbers of operations in parallel.

4.1 From Transistors to Processors

At the most basic level, modern digital hardware uses **transistors** as electronic switches. A transistor is "on" (conducting) or "off" (not conducting) depending on the voltage at its control terminal, and these two states represent binary 1 and 0. Modern chips use CMOS technology, which pairs two types of transistors so that the circuit draws significant current only while it is switching. Groups of transistors form **logic gates**, logic gates combine to build adders, multiplexers, and registers, and these in turn make up the arithmetic logic unit (ALU), the control unit, and finally the complete processor. A modern CPU contains billions of transistors.

A	B	AND	OR	NAND	XOR
0	0	0	0	1	0
0	1	0	1	1	1
1	0	0	1	1	1
1	1	1	1	0	0

Table: Truth tables of common logic gates. A half adder, the simplest circuit that adds two bits, is built from just two gates: Sum = A XOR B and Carry = A AND B. Chaining adders together gives the circuit that performs addition in a CPU.

4.2 Power, Heat, and Performance

The dynamic power consumed by a digital chip is approximately:

$$P \approx \alpha \times C \times V^2 \times f$$

where α is the fraction of gates switching, C is the switched capacitance, V is the supply voltage, and f is the clock frequency. This single formula explains many design decisions. Because power depends on the *square* of voltage, lowering the supply voltage by 10% cuts dynamic power by about 19% ($0.9^2 = 0.81$). It also explains why chip makers stopped increasing clock speeds around the mid-2000s and moved to multi-core designs: higher frequency means more power and more heat than cooling can remove. Techniques such as dynamic voltage and frequency scaling (DVFS), clock gating, and power gating are all circuit-level ideas used by the operating system and hardware to save energy.

4.3 Memory and Signal Integrity

SRAM, used for CPU caches, stores each bit in a small circuit of transistors that holds its state while powered. DRAM stores each bit as charge in a tiny capacitor, which leaks and must be refreshed thousands of times per second. Storage devices such as SSDs trap charge in flash cells. At high speed, signals travelling along motherboard tracks suffer from resistance, capacitance, and noise, so

designers use careful routing, termination, and shielding. A CSE student who understands these effects can better appreciate why memory speed, bus width, and cable length matter.

5. Electrical Circuits and Artificial Intelligence (AI)

AI applications such as image recognition, speech processing, recommendation systems, and generative AI require enormous amounts of computation. Training a modern neural network involves billions or trillions of multiply-and-add operations. These tasks run on CPUs, GPUs, and specialised AI accelerators such as tensor processing units (TPUs) and neural processing units (NPUs). All of them are built from electronic circuits that carry out calculations and move data between processing units and memory.

Circuit design influences three things that matter greatly in AI: **speed** (how many operations per second), **power consumption** (how much electricity is needed), and **heat** (how much cooling is required). In many AI systems, moving data between memory and the processor costs more energy than the arithmetic itself, which is why accelerators place memory close to the compute units and use low-precision number formats. Large data centres that train AI models consume megawatts of power, so energy-efficient hardware is both an environmental and an economic priority. On the other end of the scale, a smartphone or a camera must run AI models on a small battery, so circuit efficiency decides what is practical.

As AI systems become more widely used, engineers must consider not only algorithms and software, but also the physical computing systems that run them. A basic understanding of electrical circuits gives CSE students the context needed to evaluate hardware choices, optimise models for devices with limited power, and take part in the design of efficient AI systems.

6. Electrical Circuits in Robotics and IoT

Robotics and IoT combine computing with the physical environment. The basic structure is a loop: **sense → process → act**. A robot may use sensors to measure distance, light, or temperature; a microcontroller reads these signals through an analogue-to-digital converter (ADC) and runs a program; and motor-driver circuits control the movement of motors. The program determines what the robot should do, while the electronic circuits make it possible to sense and act.

6.1 Circuit Concepts Used in Practice

- **Voltage dividers and ADCs:** convert the changing resistance of a light or temperature sensor into a voltage that the microcontroller can measure.
- **Pull-up and pull-down resistors:** keep an input pin at a defined logic level so that a push-button does not give random readings.
- **PWM (pulse-width modulation):** switches a signal on and off rapidly to control motor speed or LED brightness without wasting energy as heat.
- **Transistor or driver circuits:** a microcontroller pin can supply only a few milliamperes, so a transistor or motor driver is needed to switch larger currents.
- **Flyback diodes:** protect electronics from the voltage spike produced when a motor or relay coil is switched off.

- **Power budgeting:** adding up the current drawn by every part to choose a suitable battery or supply.

6.2 Examples

A smart-home temperature monitor reads a sensor, converts the measurement into digital data, and sends it over Wi-Fi to a phone application. A smart irrigation system uses soil-moisture readings to switch a water pump through a relay or transistor only when the soil is dry, saving water. A line-following robot reads infrared sensors and adjusts the speed of two motors using PWM. In battery-powered IoT devices, engineers use sleep modes so that a sensor node can run for months or years on a small battery. Each example requires CSE knowledge (programming, networking, data handling) and circuit knowledge (sensing, switching, power) working together.

7. Electrical Circuits in Electric Vehicles and Renewable Energy

Electric vehicles (EVs) rely on electronic circuits for battery monitoring, motor control, charging, and safety functions. A battery-management system (BMS) measures values such as cell voltage, current, and temperature to keep the battery within safe limits, estimate the state of charge, and balance the cells. Power-electronic circuits (inverters and converters) control the flow of energy between the battery, the motor, and other vehicle systems, converting DC from the battery into the AC needed by many electric motors. Modern vehicles also contain dozens of embedded controllers that communicate over in-vehicle networks such as the CAN bus.

Renewable-energy technologies depend on circuits too. Solar photovoltaic panels produce DC electricity whose voltage and current vary with sunlight; power converters, maximum-power-point-tracking (MPPT) controllers, and inverters make that energy suitable for batteries, homes, or the power grid. Smart grids add sensors and communication so that supply and demand can be balanced in real time.

CSE engineers contribute by developing monitoring software, embedded controllers, data systems, machine-learning models for energy forecasting, and intelligent energy-management applications. These fields show how computing and electrical engineering increasingly overlap, which is especially relevant for countries such as Bangladesh that are expanding solar energy and electrified transport.

8. Future Scope: Smart Chips, Edge Computing, and Connected Systems

The future of computing is moving toward devices that are smaller, faster, more connected, and more energy efficient. **Smart chips** and specialised processors are designed for particular tasks, including AI inference, signal processing, and security. **Edge computing** allows data to be processed close to the sensor or device instead of sending everything to a distant server. This reduces delay (latency) and network use, and it improves privacy, which is especially valuable in robotics, healthcare wearables, smart cities, and industrial monitoring.

Other developing directions include low-power chips for billions of IoT devices, 5G and 6G connected systems, energy harvesting that powers small sensors from light or vibration, neuromorphic and in-memory computing that reduce data movement, and the long-term research area of quantum computing, which also depends on carefully engineered electronic control circuits. All of these developments still rely on circuits that carry signals, process data, and manage power.

CSE students who understand circuit fundamentals can communicate more effectively with hardware teams and make better design decisions when developing embedded, intelligent, or connected products. Electrical Circuit is therefore a foundation for learning, not a subject separate from modern computer science.

9. Connecting the Course to Career Opportunities

Career area	Circuit concepts that help
Embedded and IoT developer	Voltage levels, sensors, ADC, power budgeting, protection circuits
Robotics engineer	Motor drivers, PWM, feedback, battery systems
Hardware / computer architecture roles	Logic gates, timing, power and heat, signal integrity
AI hardware and optimisation	Power-efficiency, accelerators, memory energy cost
Network and systems engineer	Signal transmission, power supplies, troubleshooting
Energy and EV software	Battery management, converters, monitoring systems

10. Conclusion

Electrical Circuit is an important course for Computer Science Engineering students because it explains the electronic foundation of computing devices. Its concepts support the study of computer hardware, digital logic, AI processors, robotics, IoT, electric vehicles, and renewable-energy systems. Circuit knowledge also helps students understand the relationship between hardware performance, power consumption, and software behaviour, and it develops problem-solving habits such as breaking a system into parts, modelling it with equations, and testing it carefully.

As technology continues to develop, CSE engineers will increasingly work with systems that combine software, electronics, sensors, and intelligent processing. Learning electrical circuits gives students a practical foundation for understanding these systems and for preparing for future opportunities in computer science and engineering.

11. References

The following learning resources provide background information on circuit fundamentals, digital electronics, and computer hardware:

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